MACHINE LEARNING PROJECTS / SPRING 2026

NYC PLUTO: Building Typologies Across the Five Boroughs

Unsupervised segmentation of the New York City building stock

RESEARCH QUESTION

Using only the physical and functional characteristics of NYC buildings, what typologies emerge across the five boroughs?

Borough and land use are deliberately withheld from the analysis. *Any geographic pattern that emerges is therefore a discovery, not a built-in assumption.*

THE APPROACH

Unsupervised Clustering

- Clustering groups buildings with similar measurable profiles, without using any predefined label.
- We let the data reveal recurring building types, then ask how those types map onto the city.
- No target variable, no borough, no land use: only what each building physically is.

Same points, no labels on the left; groups discovered from structure on the right.

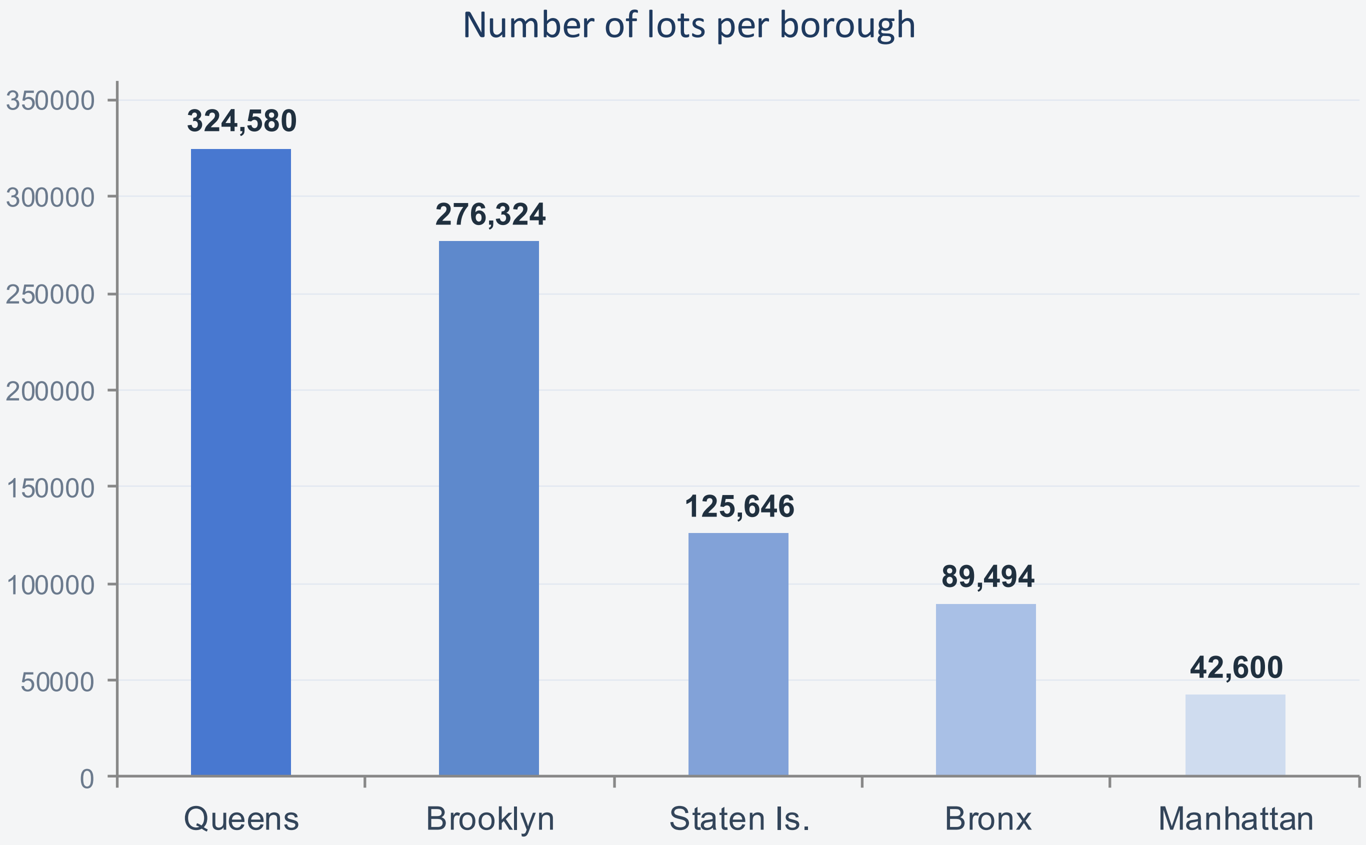
THE DATASET

NYC PLUTO, Before Any Processing

858,644
tax lots (rows)

101
columns

- One row = one tax lot, the smallest unit of NYC real property.
- Source: NYC Department of City Planning, Open Data.
- Highly imbalanced: Brooklyn and Queens alone are 70% of all lots.



PIPELINE

From Raw Data to Clusters

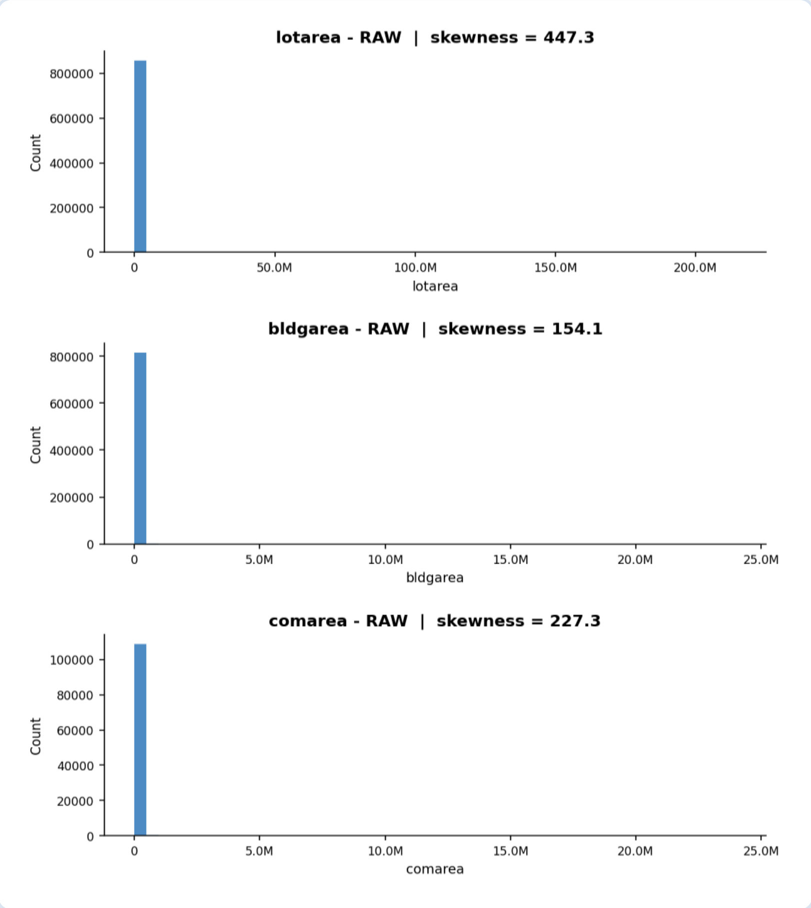
1. **Raw PLUTO**
   858,644 x 101
2. **Clean**
   impossible values, IQR cap
3. **Engineer**
   3 new features
4. **Scale**
   log1p + StandardScaler
5. **X_scaled**
   682,515 x 16
6. **PCA**
   10 comp, 92% var
7. **Cluster + Evaluate**
   K-Means, DBSCAN, Ward

Three Problems the EDA Revealed

1 **Extreme right skewness**
Lot area skewness = 447: a few giant lots crush everything into one bar.

2 **Structurally impossible values**
yearbuilt = 0 on 40,115 lots, bldgarea = 0 on 41,109 lots: present but invalid.

3 **Missing values**
numfloors absent on 8.09% of lots, area features on 5.41%: no recorded data.

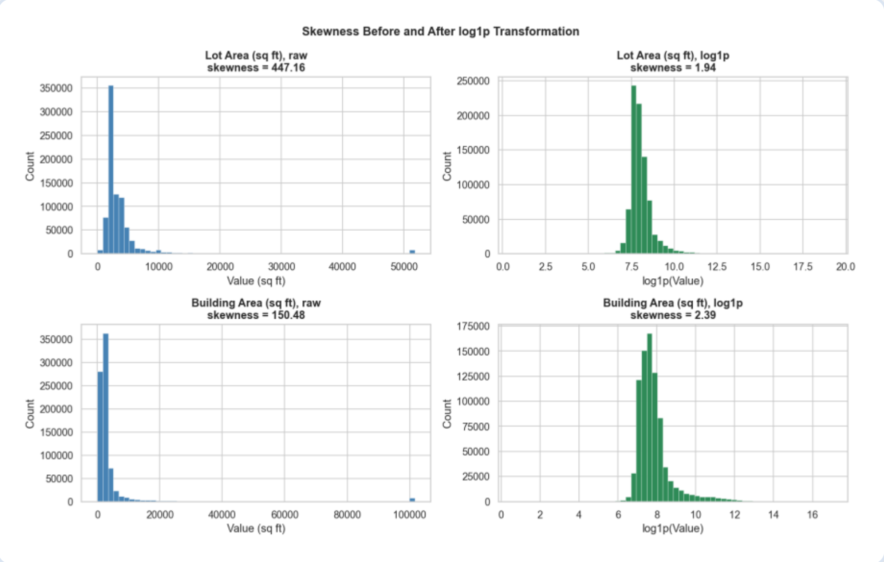


Raw distributions: nearly all mass sits in a single bar.

PREPROCESSING

From Problems to a Clean Feature Matrix

Skewness	log1p on 11 features (447 to about 2)
Impossible values	set to NaN (year built, building area, floors)
Missing values	drop rows missing any clustering feature
Heavy-tailed outliers	IQR winsorization (1.5 x IQR) on 7 features



Skewness collapses after log1p.

Engineered features (3): building_age = 2026 - yearbuilt building_density = bldgarea / lotarea residential_ratio = unitsres / unitstotal

OUTPUT OF PREPROCESSING

The Result: X_scaled

682,515	16	0 / 1	20.5%
tax lots	features	mean / std	rows removed

Every feature is standardized to mean 0 and standard deviation 1, so no single feature can dominate the distance. The 176,129 removed lots were mostly missing year built or building area.

FEATURE SELECTION

16 Features for Clustering, 5 Held Out for Interpretation

7 Physical features

lot area, building area, floors, lot frontage, lot depth, building age, building density

9 Functional features

total units, residential units, residential area, commercial area, office area, retail area, garage area, storage area, residential ratio

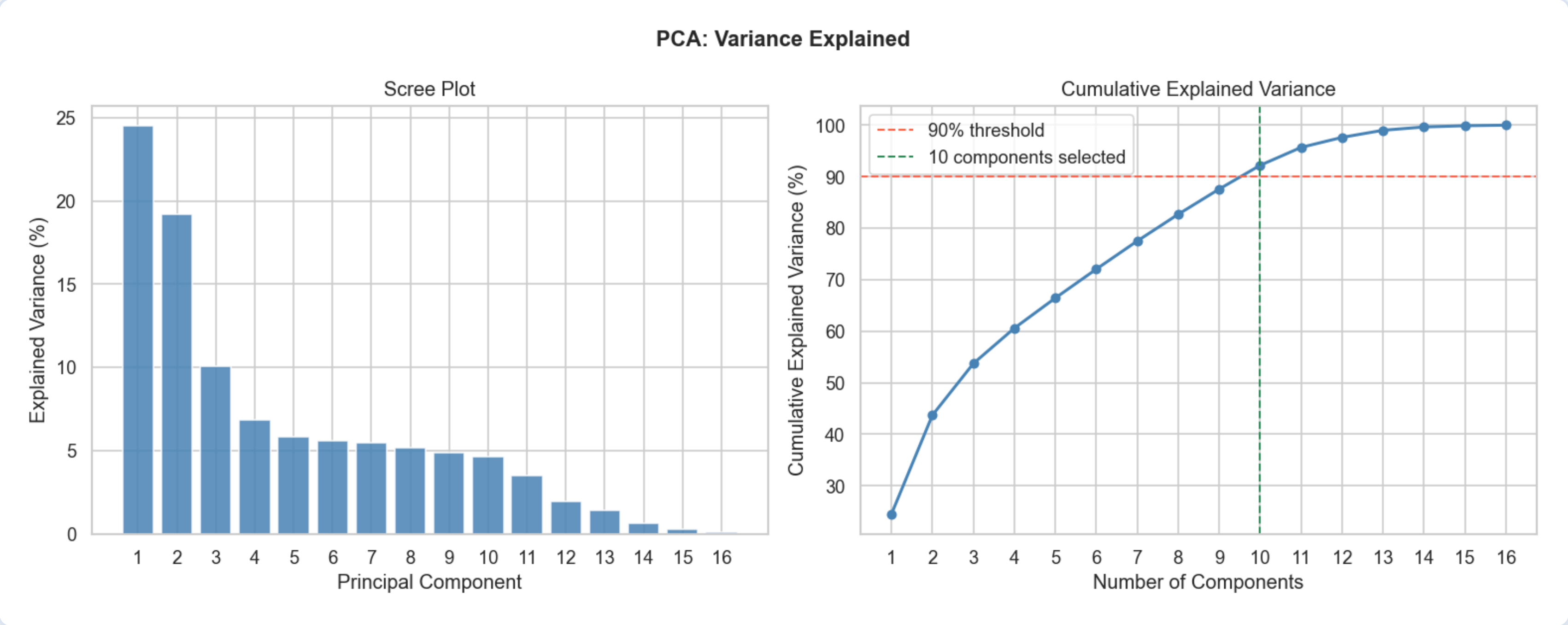
16 features enter clustering. 5 attributes are held out for interpretation only: borough, land use, latitude, longitude, assessed value.
Withheld so clusters cannot simply reproduce administrative categories; their match is checked afterward as a blind validation.

METHODS

Unsupervised Clustering

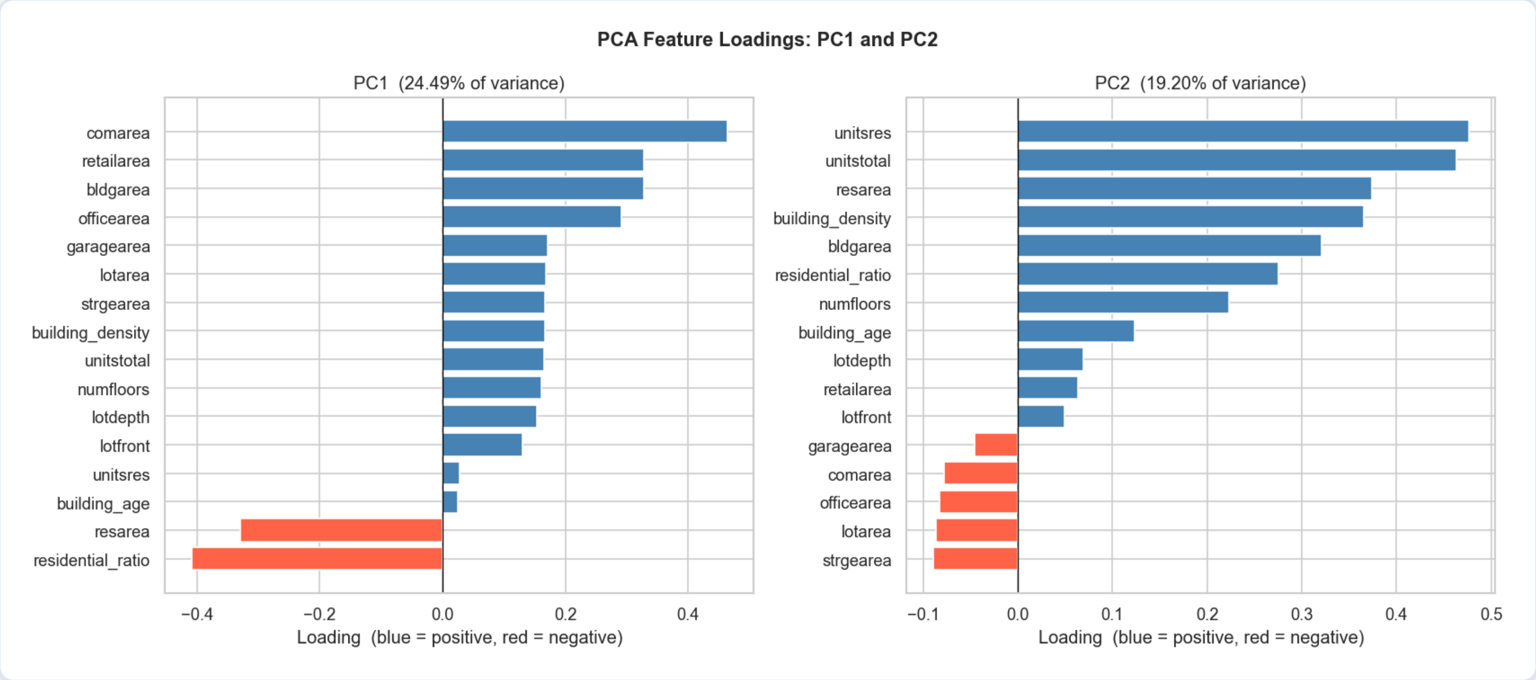
PCA, K-Means, DBSCAN, and a hierarchical cross-check

Why PCA Before Clustering, Not Only After



- Standard course pipeline: standardize, then reduce, then cluster.
- Decorrelation: PCA removes the repeated information that distorts the density estimate DBSCAN relies on, and it preserves global geometry, so it is the clustering input.
- 10 components retain 92.19% of total variance; t-SNE is kept for visualization only, never as an input.

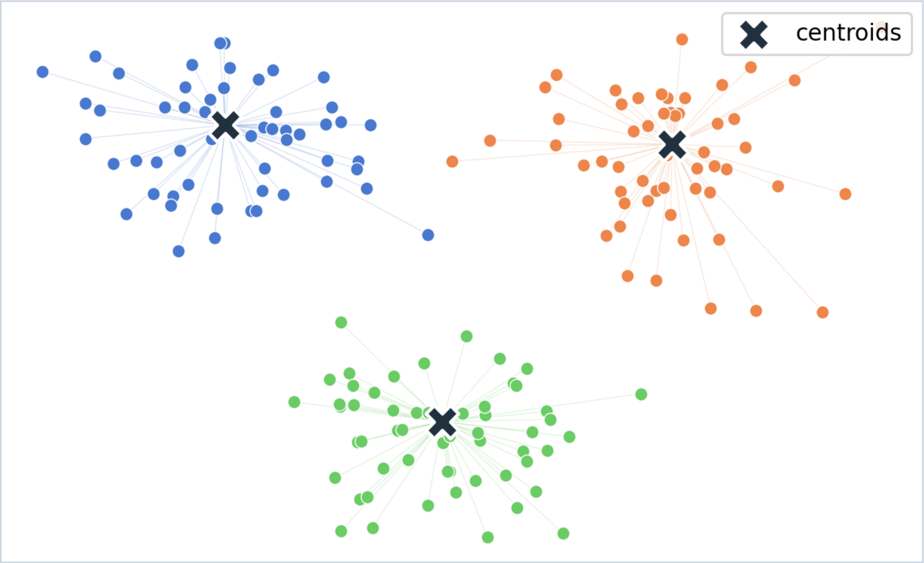
What the First Two PCA Axes Represent



- PC1 (24.49%) is a use-mix axis: commercial, retail and size load positive, residential negative (housing-heavy lots score low); it runs from housing to commercial use.
- PC2 (19.20%) is a residential-intensity axis: units, residential area, density and floors rise together.
- Clusters are interpreted on the original features, since each component mixes all 16.

K-Means: The Primary Algorithm

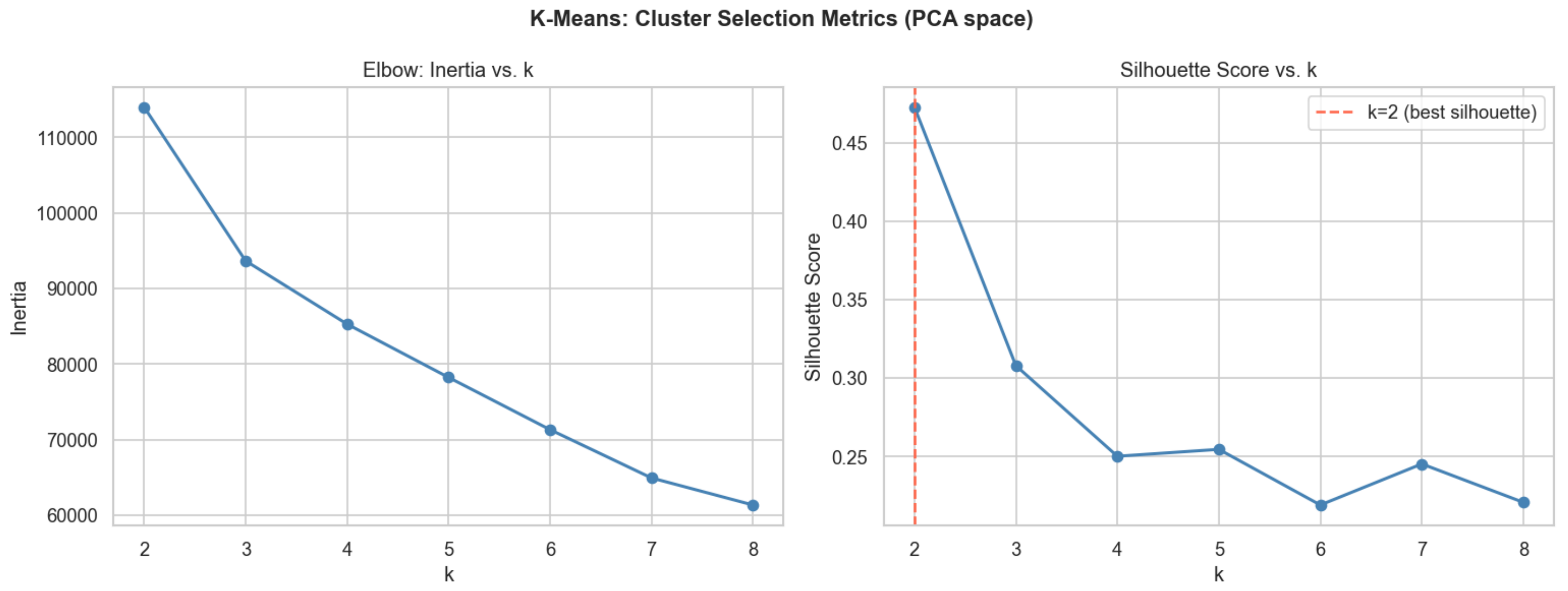
Each lot joins its nearest centroid;
centroids minimize within-cluster distance



- Partitions every lot into a fixed number k of groups by minimizing the distance to each group centroid (within-cluster variance).
- Strengths: it scales to all 682,515 lots, the partition is stable and reproducible, and each cluster is summarized by an interpretable centroid (its mean building profile).
- Limits: k must be chosen in advance, and it assumes roughly convex groups.
- Two other algorithms make up for these limits: DBSCAN (no preset k, any shape) and Ward (hierarchical).

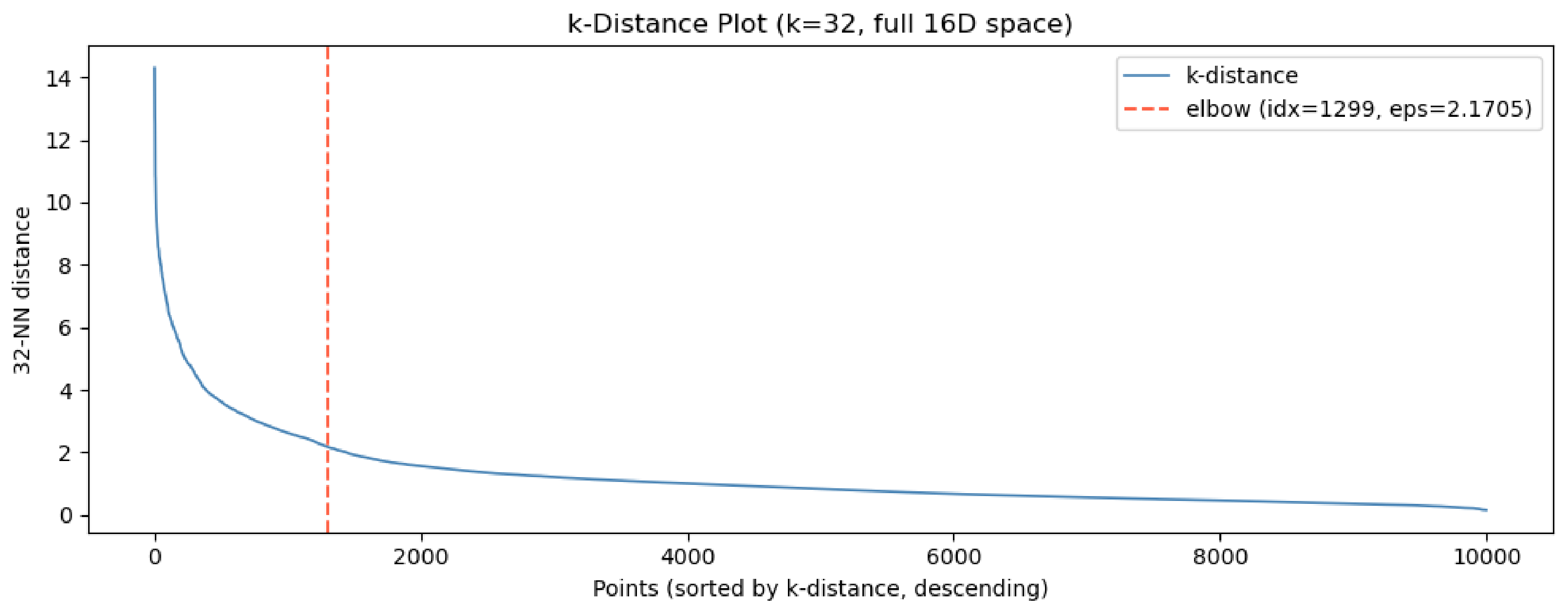
CHOOSING k

Silhouette Picks k=2, But k=2 Is Too Coarse



- Rule fixed in advance: choose the k that maximizes the silhouette in PCA space.
- k=2 wins (silhouette 0.4734): the lots split 85% / 15% into a large residential group and a smaller non-residential group, the robust primary result.
- k=2 is too coarse for typology, so we also report k=5, whose silhouette is far lower; its value is interpretability, not the metric.
- Why k=5 and not k=3 or k=4: k=3 stays too coarse to separate the five typologies, and among the finer cuts k=5 has the best silhouette (just above k=4 and k=6), so it is the natural granularity for interpretation.

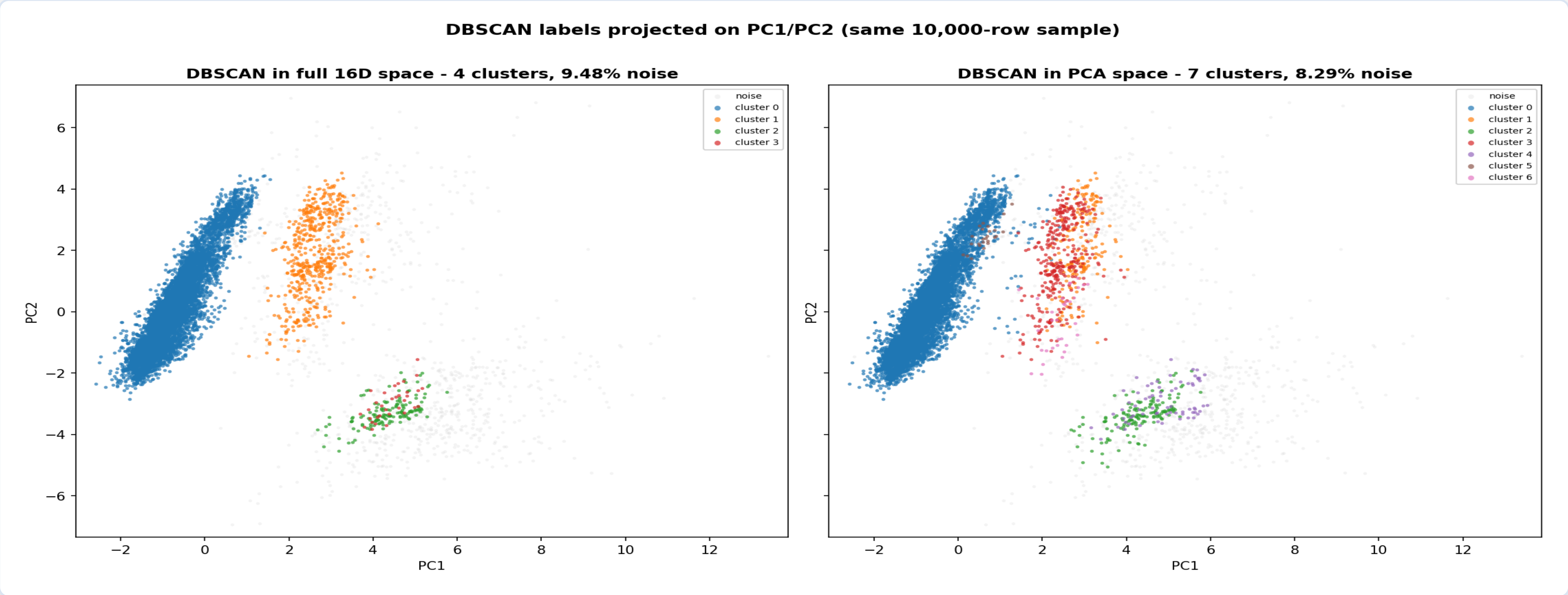
DBSCAN: A Density-Based Algorithm



- Density-based: it finds dense regions of any shape, labels sparse points as noise, and does not need k in advance.
- eps is read from the full-space k-distance elbow shown above (eps = 2.1705), so the parameter is justified, not arbitrary.
- Full 16D space: 4 clusters, 9.48% noise; non-noise labels match K-Means k=2 at ARI 0.9904 (full-space agreement; the PCA-space value is on the evaluation slide).

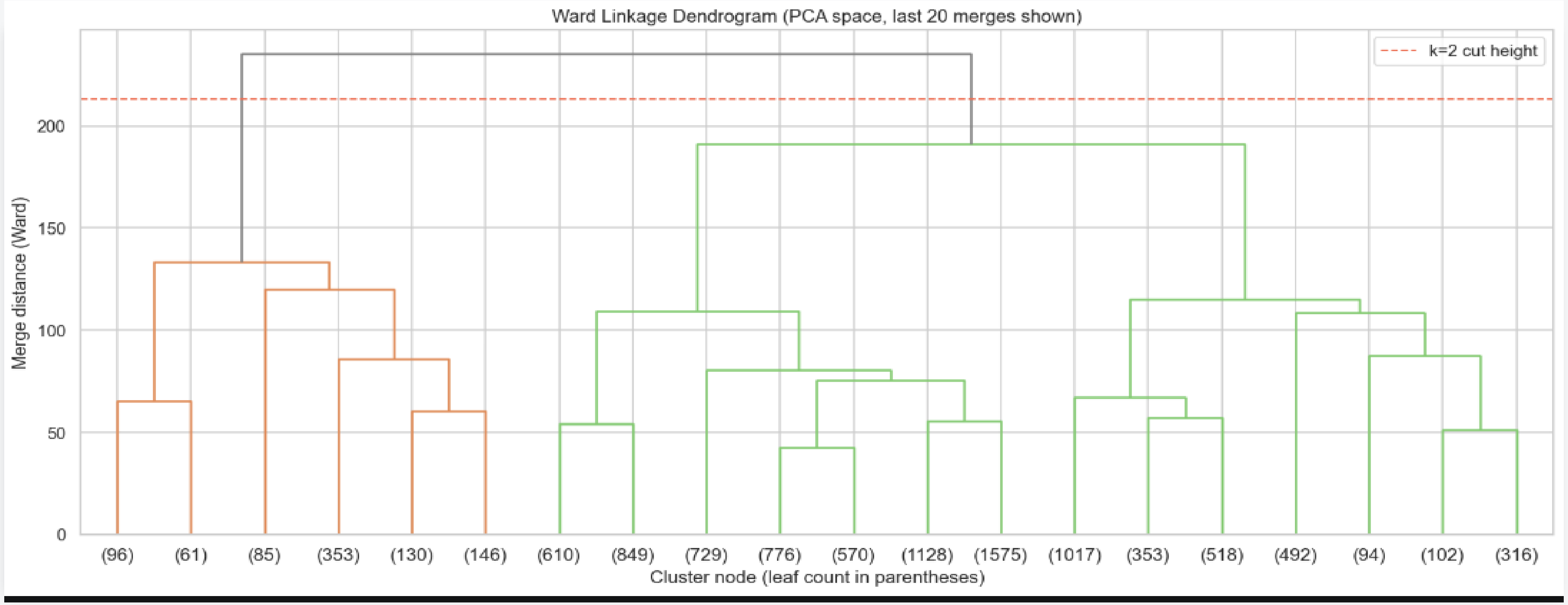
CLUSTERING

In PCA Space, the Non-Residential Minority Splits into Distinct Dense Groups



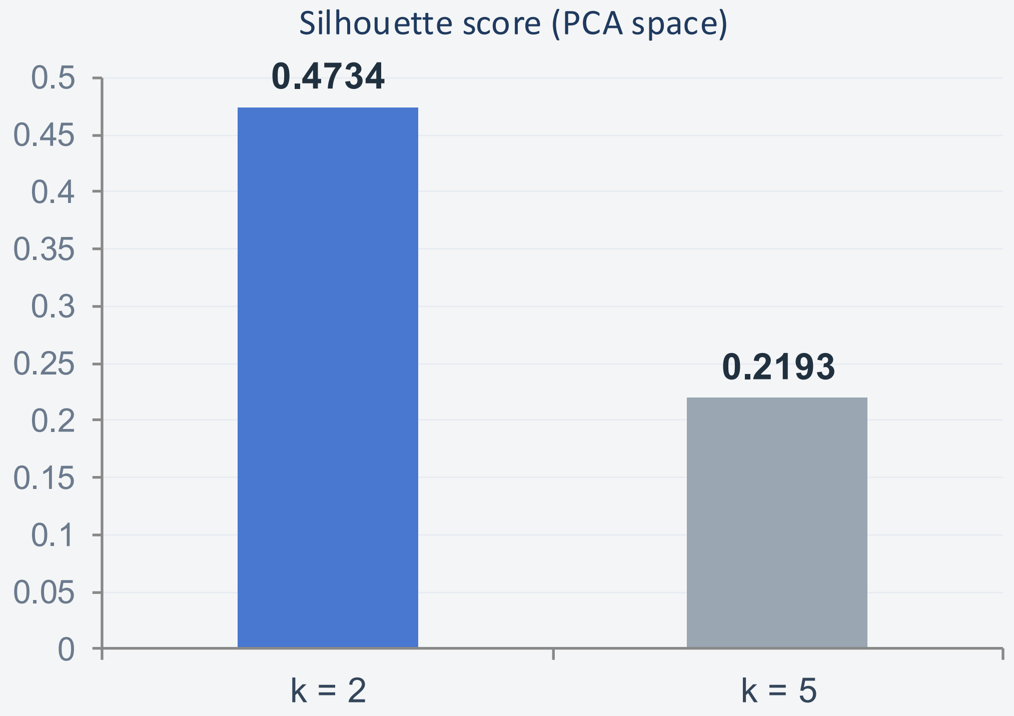
Full space: correlated features blur the local density, so the non-residential minority stays in a few blobs. PCA space: one dominant residential core plus several small dense pockets (7 clusters, 8.29% noise, comparable to 9.48%). *Caveat: the pocket count shifts with eps (sensitivity grid 7, 7, 4), so it is indicative, not a definitive count.*

Hierarchical Clustering: Seeing the Nesting



- Ward linkage merges the pair of clusters that least increases within-cluster variance, the same objective family as K-Means.
- The highest merge separates residential from non-residential (the k=2 split); each branch then subdivides into the finer typologies (k=5).
- Used only as a visualization aid to justify the move from k=2 to k=5, not as a third decision algorithm.

The Main Split Is Robust



Silhouette computed on a 10,000-lot sample.

- K-Means vs DBSCAN (PCA space): ARI 0.9587 on non-noise lots, a high agreement from an independent density-based method.
- Ward recovers the same two-level nesting in the dendrogram, corroborating the split and the move to five typologies.
- Methods with different objectives converge on the same primary division. That convergence is the main evaluation result.

ARI 0.9587

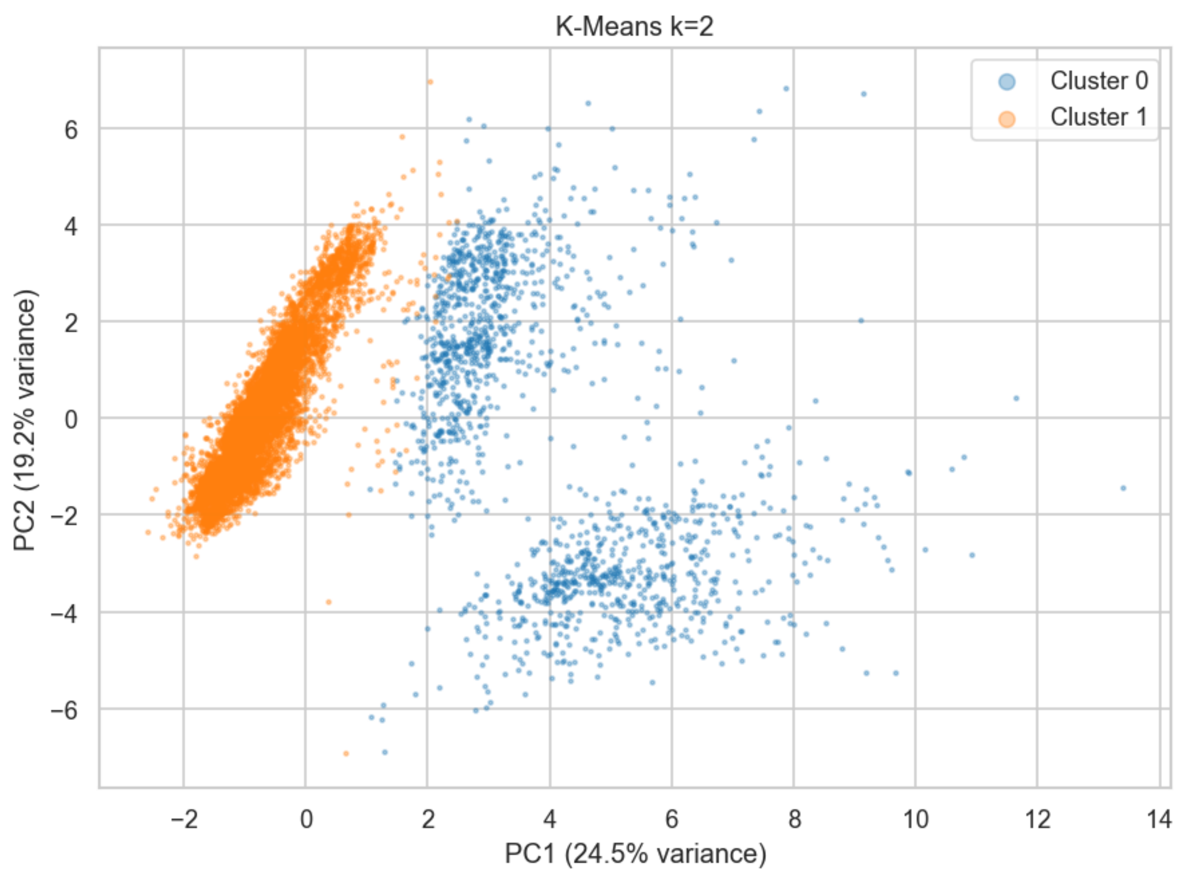
K-Means and DBSCAN agree on the primary split

VISUALIZATION

Seeing the Clusters

PCA and t-SNE views of the five typologies

The Primary Split, Seen



Real notebook output · 10,000-lot sample · clustering in full 10-D PCA space.

Two groups, one clean cut.

PC1 use-mix:
residential to commercial

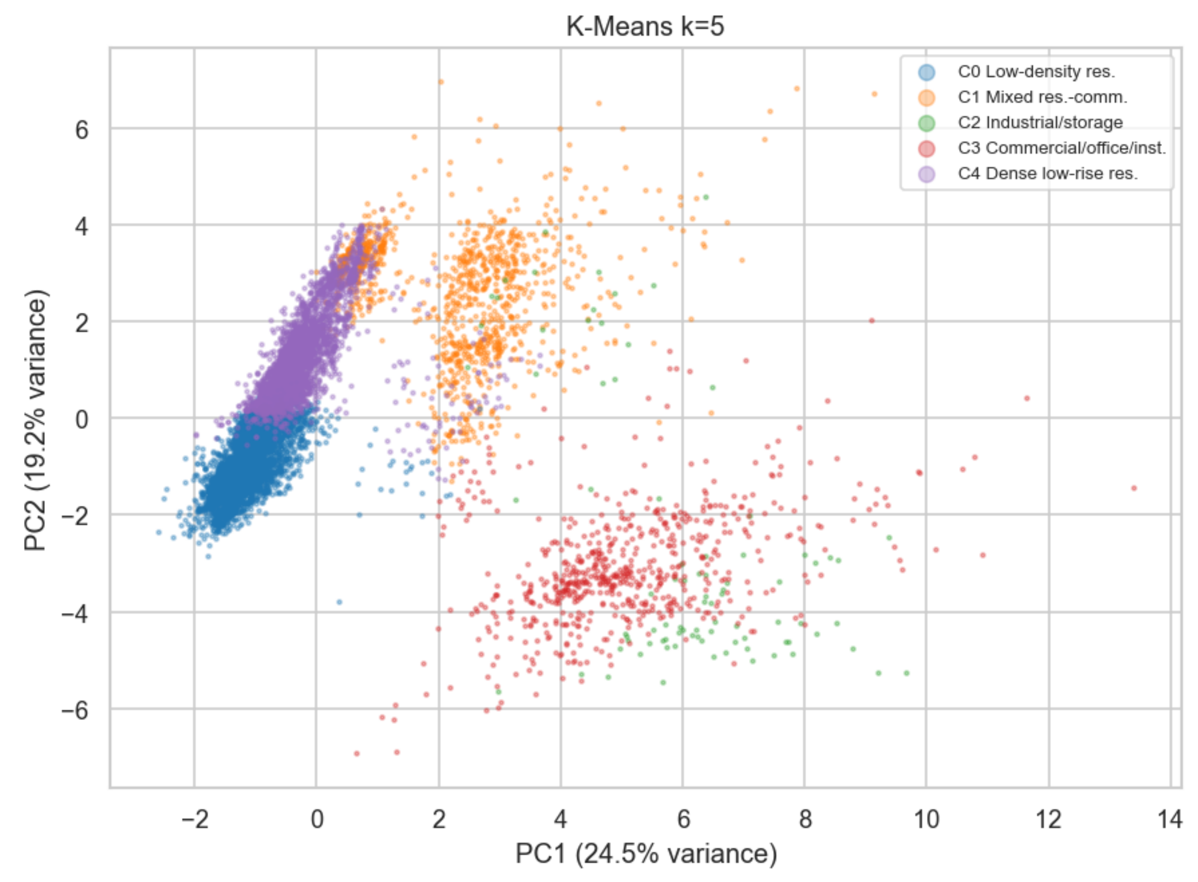
PC2 residential intensity:
density, units, floors increase

- Cloud splits cleanly along PC1
- Cluster 0: residential, 85%
- Cluster 1: non-residential, 15%
- The robust k=2 result, now visible

Apparent sub-groups = 2-D projection effect. k-Means draws one boundary in full 10-D space; they resolve into the k=5 typologies.

VISUALIZATION

Five Typologies Emerge



Real notebook output · same 10,000-lot sample, colored by K-Means k=5.

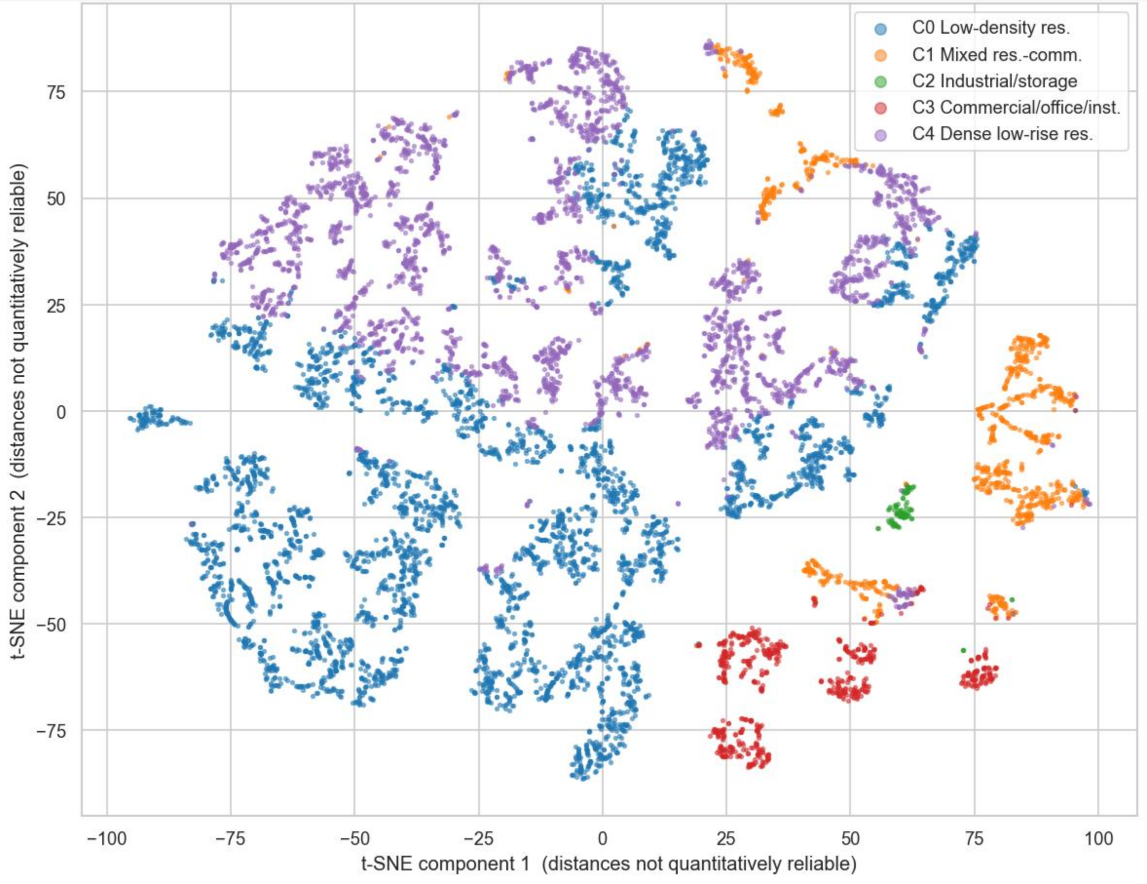
One residential mass, five typologies.

- C0: Low-density residential · 49%
- C4: Dense low-rise residential · 33%
- C1: Mixed residential-commercial · 10%
- C3: Commercial / office / institutional · 7%
- C2: Industrial / storage · 1%

A continuum, not boxes: residential to mixed to commercial to industrial along PC1.

k=5 chosen for interpretability, not the metric. Silhouette 0.22 vs 0.47 at k=2, so k=2 stays the optimum. Overlap on screen = projection (44% of variance).

t-SNE: A Second View



Real notebook output · t-SNE of the 10,000-lot PCA-space sample (perplexity 30), colored by K-Means k=5.

A non-linear view: the typologies hold.

- Residential (C0, C4) fills the left
- Mixed-use (C1) gathers on the right
- Commercial / institutional (C3) at the bottom
- Industrial / storage (C2) sits clearly apart
- **Typologies form coherent regions**

Visualization only. Course pipeline: never a clustering input. We cluster in PCA space, then color this embedding. Preserves local neighborhoods, so distances, gaps and sizes on screen are not reliable.

CLUSTERING RESULT

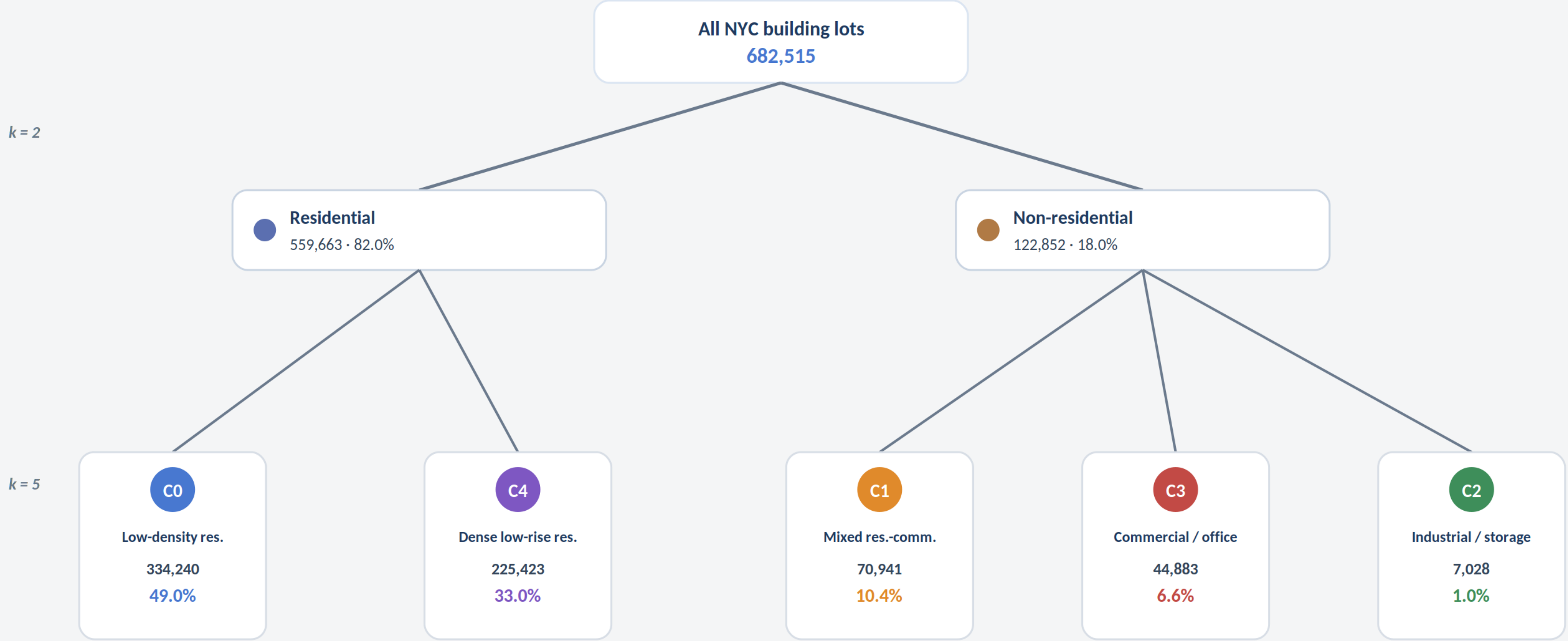
The Five Building Typologies

What each cluster is, and what sets it apart

CLUSTERING RESULT

One City, Two Levels, Five Types

The whole building stock splits in two, then each side resolves into its typologies.



Real notebook output · cluster sizes from K-Means on all 682,515 lots · nesting confirmed by the Ward dendrogram.

BLIND VALIDATION

Did the City Come Back?

Borough was never given to the algorithm, so let's check

BLIND VALIDATION

The Geography Was Never an Input

Cluster only on building form, yet the typologies rebuild the map of New York.

		C0 Low-density residential	C4 Dense low-rise residential	C1 Mixed res.-comm.	C3 Commercial office/inst.	C2 Industrial storage
Manhattan	MN	0.3	31.3	47.7	17.8	2.9
Brooklyn	BK	30.5	51.3	11.3	6.1	0.9
Queens	QN	62.7	24.2	6.4	5.7	1.1
The Bronx	BX	39.7	38.6	11.8	8.8	1.2
Staten Island	SI	90.5	3.6	1.6	3.8	0.4

% within borough

each row sums to 100%

Staten Island
90% low-density, pure suburbia

Queens
63% low-density, sprawling residential

Brooklyn
51% dense low-rise, the brownstone belt

The Bronx
split evenly between the two

Manhattan
≈0% low-density · 48% mixed-use

Real notebook output · typology share within each borough (rows sum to 100%) · borough label withheld from clustering, used here only to check.

BLIND VALIDATION

Independent Sources Agree

Official NYC data shows the same borough pattern: our blind clustering matched reality.

"In Manhattan, nearly all units (97%) are in buildings of six or more units, while in Staten Island, 83% of units are in one- or two-family homes."

[1] CHPC New York, Taking Stock (NYC housing stock)

"Mixed-use share by borough: Manhattan 33.2%, Brooklyn 9.0%, the Bronx 5.2%, Queens 4.2%, Staten Island 1.4%."

[2] PLUTO-based analysis, NYC Dept. of City Planning data

	Low-density / small homes		Mixed-use	
	Ours (C0)	Official[1]	Ours (C1)	Official[2]
Staten Island	90.5%	83%	1.6%	1.4%
Queens	62.7%	high	6.4%	4.2%
Brooklyn	30.5%	mixed	11.3%	9.0%
The Bronx	39.7%	mixed	11.8%	5.2%
Manhattan	0.3%	~3%	47.7%	33.2%

Note: official figures use different counting methods, so absolute values differ, but the relative ranking across boroughs ma tches our typologies. That agreement is the validation.

CONCLUSION

Back to the Question

What building form alone can reveal about a city

CONCLUSION

What We Found

Building form alone organizes all of New York, and we checked it four ways.

1 A two-level structure

One robust split: residential vs non-residential (82/18), nesting into five interpretable typologies.

2 Three methods converge

K-Means, DBSCAN and Ward independently recover the same primary division (ARI ≈ 0.96).

3 Typologies match official land use

Each typology is dominated by the official land-use code it should be, though land use was never an input.

4 The geography returns

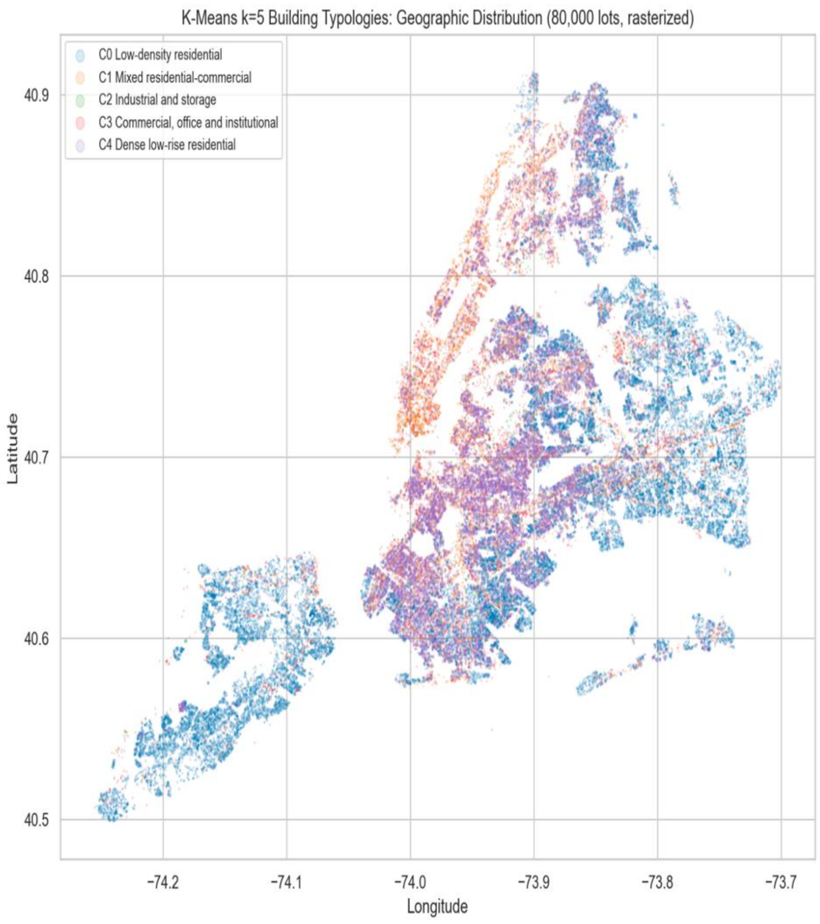
Borough was withheld, yet the typologies rebuild NYC's real map, confirmed by independent official data.

The Answer

We blindfolded the algorithm, hid borough and land use, and it still rebuilt the city.

The physical and functional character of a lot is enough to recover the urban geography of New York, both by function and by place.

הפיזור הגאוגרפי של הטיפולוגיות, משוחזר מצורת הבניין בלבד וללא borough כקלט. הוא משחזר את מפת ניו יורק



OPENING

Beyond New York

If building form alone can rebuild New York, the same method can read any city it has never seen.

This question now extends into Barzel Analytics, a real-estate analytics venture that turns raw property data into actionable market intelligence.